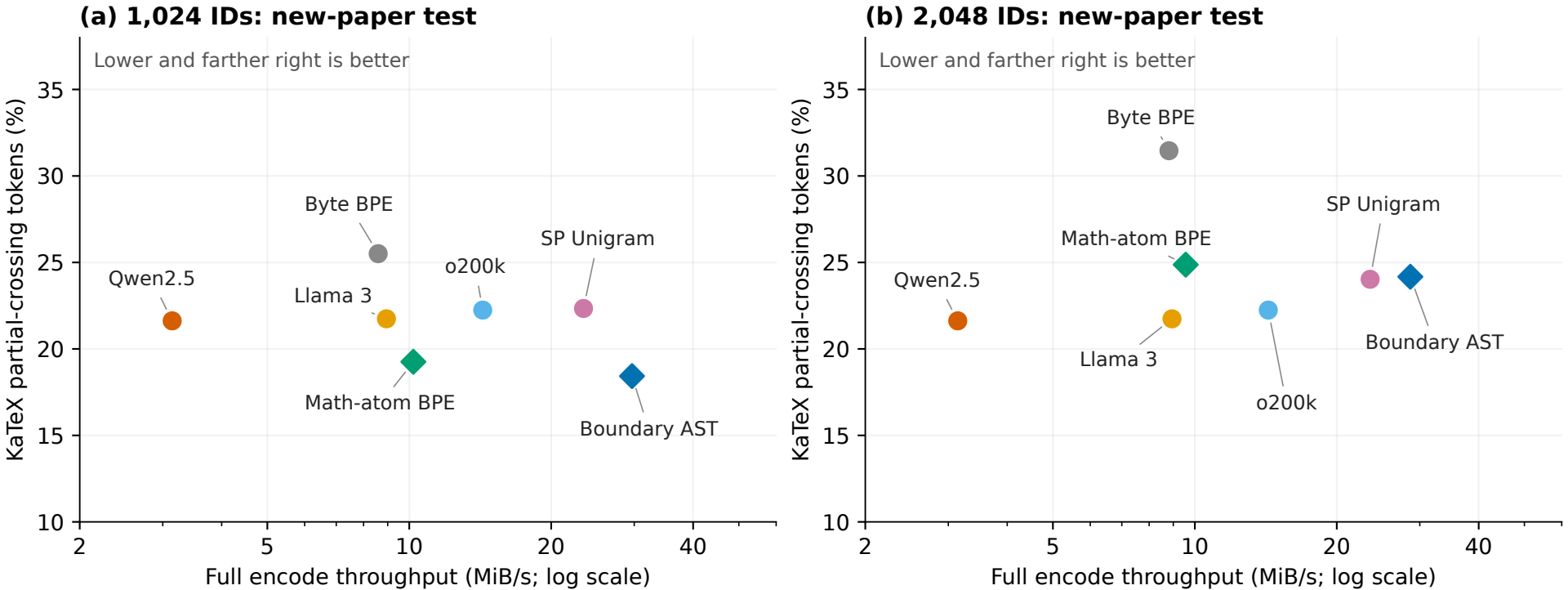
(a) 1,024 IDs: new-paper test
Lower and farther right is better
KaTeX partial-crossing tokens (%)
Full encode throughput (MiB/s; log scale)
Qwen2.5
Byte BPE
Llama 3
o200k
SP Unigram
Math-atom BPE
Boundary AST
(b) 2,048 IDs: new-paper test
Lower and farther right is better
KaTeX partial-crossing tokens (%)
Full encode throughput (MiB/s; log scale)
Byte BPE
SP Unigram
Math-atom BPE
Qwen2.5
Llama 3
o200k
Boundary AST